

The Sraffian Economy as an Autocatalytic Set: Catalytic Closure, Phase Transitions, and Economic Crisis

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We identify Sraffa’s classical production system as an autocatalytic set (ACS) in the sense of Jain and Krishna [*Phys. Rev. Lett.* **81**, 5684 (1998)]. The $n \times n$ input-output matrix A is the adjacency matrix of the production network: Sraffa’s *basic commodities* — those entering directly or indirectly into all other production processes — form the dominant ACS, satisfying the catalytic closure condition for the economy’s self-reproduction. By the Perron-Frobenius theorem, the asymptotically stable attractor of the competitive production dynamics is the leading eigenvector of A — precisely Sraffa’s *standard commodity*, Ricardo’s invariable measure of value. The maximum rate of profit $R = 1/\lambda_1 - 1$, where λ_1 is the dominant eigenvalue, measures the ACS’s net catalytic surplus above its own reproduction requirements. Disruption of a *keystone sector* — a basic commodity node with high out-degree in the production graph — severs catalytic links to multiple downstream sectors simultaneously, producing a discontinuous collapse in ACS connectivity: a phase transition rather than a smooth departure from equilibrium. We show this discontinuity is structurally distinct from the smooth Hamiltonian non-conservation of expanded reproduction, providing a precise mechanism for why economic crises are discontinuous. The framework connects Sraffian production theory, ACS theory, and the thermodynamics of self-reproducing networks.

1 Introduction

Autocatalytic sets (ACS) — subgraphs of a directed interaction network in which every node has at least one incoming catalytic link from within the same subgraph — were introduced by Jain and Krishna as a model for the emergence of self-reproducing chemical networks (Jain and Krishna 1998, 2001). The central result is that the dominant ACS of a random catalytic network evolves inevitably toward higher connectivity and complexity, with its population distribution converging to the leading eigenvector of the catalytic adjacency matrix. Phase transitions occur when keystone nodes — those with high out-degree in the dominant ACS — are disrupted, producing discontinuous collapses in network size that recover through reconstitution around a new attractor.

This paper identifies an exact structural correspondence between the ACS framework and Sraffa’s classical production system (Sraffa 1960). The correspondence runs deeper than analogy: the input-output matrix of a Sraffian economy *is* the adjacency matrix of a directed catalytic network, and the properties of basic commodities, the standard commodity, and the maximum rate of profit follow directly from the Perron-Frobenius theorem applied to this ACS. The result provides a rigorous network-theoretic foundation for classical production theory and a precise account of economic crisis as a phase transition in the production network’s ACS structure.

The connection between ACS theory and economic systems has been explored in the context of technological evolution (Hordijk, Steel, and Kauffman 2022), and the Perron-Frobenius structure of Sraffa-Leontief systems has been studied extensively (Emmenegger, Falk, and Schefold 2020). The present paper combines these threads: identifying Sraffa’s *basic commodities* with the dominant ACS, the *standard commodity* with the Jain-Krishna attractor, and *economic crisis* with the phase transition triggered by keystone sector disruption.

2 The Sraffian Production System as a Directed Catalytic Network

2.1 Setup

Consider an economy with n sectors of production. Each sector i produces a single commodity using inputs from other sectors and labour. The technology is summarised by:

- A : the $n \times n$ matrix of direct input-output coefficients, where $A_{ij} > 0$ if one unit of sector j ’s output requires A_{ij} units of sector i ’s output as input
- ℓ : the n -vector of labour coefficients, $\ell_i > 0$ the labour required per unit of output in sector i
- x : the n -vector of gross outputs

The economy is *viable* — capable of sustaining itself — if and only if the dominant eigenvalue λ_1 of A satisfies $\lambda_1 < 1$, so that some net output remains after reproduction of the inputs (Sraffa 1960).

2.2 The Input-Output Matrix as Catalytic Adjacency Matrix

Define the directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \dots, n\}$ is the set of sectors (nodes) and there is a directed edge $(i \rightarrow j)$ — sector i catalyses sector j ’s production — if and only if $A_{ij} > 0$: sector i ’s output is required as an input for sector j .

This is precisely the adjacency matrix of a catalytic network: sector i “catalyses” sector j ’s production in the sense that without sector i ’s output, sector j cannot operate. The Jain-Krishna ACS condition — every node in the set has at least one incoming catalytic link from within the set — corresponds exactly to the Sraffian notion of *basic commodities*.

2.3 Basic Commodities as the Dominant ACS

Sraffa defines a commodity as *basic* if it “enters directly or indirectly into the production of all commodities” (Sraffa 1960, 7). In graph-theoretic terms: commodity i is basic if every node $j \in \mathcal{V}$ is reachable from node i in $\mathcal{G}$.

Proposition 1. *The set $\mathcal{B}$ of basic commodities forms the dominant ACS of the production network $\mathcal{G}$.*

Proof. (i) Catalytic closure: since every basic commodity enters the production of every other commodity (directly or indirectly), every node in $\mathcal{B}$ has at least one incoming edge from within $\mathcal{B}$. (ii) Dominance: $\mathcal{B}$ contains every node reachable from every other node, making it the largest strongly connected component with universal reach. Non-basic commodities are leaves: they receive catalytic support from $\mathcal{B}$ but do not return it universally. $\square$

The non-basic sectors are analogous to “peripheral” nodes in the Jain-Krishna model: they depend on the dominant ACS for their catalytic inputs but do not provide universal catalytic support in return.

3 The Standard Commodity as ACS Attractor

3.1 Jain-Krishna Population Dynamics

Jain and Krishna study the population dynamics of an ACS under the replicator equation (Jain and Krishna 1998):

$$\dot{x}_i = \sum_j c_{ij}x_j - x_i \sum_{k,j} c_{kj}x_j, \quad i = 1, \dots, n$$

where c_{ij} is the catalytic contribution of species j to species i . By the Perron-Frobenius theorem, the asymptotically stable fixed point of this dynamics is the leading eigenvector $\mathbf{x}^* = \mathbf{v}_1$ corresponding to the dominant eigenvalue λ_1 of the catalytic matrix $C = [c_{ij}]$.

3.2 The Standard Commodity as Leading Eigenvector

In Sraffa’s production system, the competitive dynamics that distribute capital across sectors drive outputs toward the *standard commodity* — the unique output vector $\mathbf{x}^*$ such that its composition of gross outputs, net outputs, and means of production are all in the same proportions:

$$(I - A)\mathbf{x}^* = \lambda_1 A\mathbf{x}^*, \quad \mathbf{x}^* = \mathbf{v}_1(A)$$

where $\mathbf{v}_1(A)$ is the leading eigenvector of A (Sraffa 1960).

Proposition 2. *The standard commodity is the Jain-Krishna attractor of the Sraffian production dynamics under the identification $C = A$.*

This is not a formal curiosity but an economic result: the standard commodity is simultaneously (i) Ricardo’s invariable measure of value — immune to distributional changes between wages and profits, (ii) the algebraic device for deriving the price equations, and (iii) the dynamical fixed point of competitive selection in the production network. It is all three for the same reason: it is the leading eigenvector of the autocatalytic adjacency matrix.

3.3 The Maximum Rate of Profit as Net Catalytic Surplus

The maximum rate of profit in Sraffa’s system is:

$$R = \frac{1}{\lambda_1} - 1$$

where λ_1 is the dominant eigenvalue of A . Three regimes:

λ_1	R	Economic / ACS interpretation
$\lambda_1 < 1$	$R > 0$	Surplus economy — ACS has positive net catalytic throughput
$\lambda_1 = 1$	$R = 0$	Subsistence economy — ACS is self-sustaining without surplus
$\lambda_1 > 1$	$R < 0$	Sub-viable — inputs exceed outputs; the network cannot sustain itself

R is therefore a direct measure of the ACS’s **net catalytic productivity**: the surplus the self-reproducing network generates above its own reproduction requirements. The Jain-Krishna eigenvalue condition and Sraffa’s maximum rate of profit are the same object from two perspectives.

4 Phase Transitions: Keystone Disruption and Economic Crisis

4.1 Keystone Sectors

In the Jain-Krishna model, the *least-fit* node is selected for mutation or disruption. When this node is a *keystone* of the dominant ACS — a node with high out-degree, whose outputs catalyse many downstream nodes — its removal severs multiple catalytic links simultaneously (Jain and Krishna 1998, fig. 3). The result is a sudden, discontinuous collapse in ACS size: a phase transition.

Definition. Sector i is a *keystone sector* if (i) $i \in \mathcal{B}$ (it is a basic commodity) and (ii) the out-degree of node i in $\mathcal{G}$ is high relative to the mean — many sectors depend on i 's output as a direct input.

Keystone sectors in a capitalist economy include energy, transport infrastructure, basic materials, and financial intermediation: sectors whose outputs are universally required as inputs across the production network.

4.2 The Phase Transition

Proposition 3. *Disruption of a keystone sector $i \in \mathcal{B}$ with out-degree $d_i^+ \geq \theta_c$ (above a critical threshold) produces a discontinuous collapse in the size of the dominant ACS.*

The mechanism: removing node i from $\mathcal{G}$ breaks the strongly-connected-component structure of $\mathcal{B}$. Sectors that received their only path to universal catalytic reach through i are no longer basic — they become leaves or isolated. If d_i^+ is large, many sectors lose their catalytic closure simultaneously. The dominant ACS shrinks discontinuously, reducing λ_1 and potentially making the economy sub-viable ($R < 0$) in the affected subgraph.

This is structurally distinct from the smooth Hamiltonian non-conservation of expanded reproduction. The Hamiltonian framework characterises economic dynamics as $\dot{H} < 0$ during crisis — a smooth departure from the conservative manifold. The ACS framework explains *why* crises are discontinuous: they are phase transitions triggered by crossing a critical threshold in catalytic connectivity. Together the two frameworks give the complete picture — the Hamiltonian describes the conservative dynamics around the attractor; the ACS describes the phase transitions away from it.

4.3 Recovery and Reconstitution

Jain and Krishna show that after a phase transition, the system reconstitutes a new dominant ACS — typically one that incorporates the mutated or disrupted node in a new role, achieving higher connectivity than before the disruption. The economic analogue is the long-wave recovery: sufficient devaluation of the capital stock (lowering λ_1 , raising R) creates conditions for a new expansion,

typically incorporating new sectors (technologies) into the dominant ACS at higher connectivity. This formalises Schumpeter’s “creative destruction” as a precise ACS reconstitution event.

5 Discussion

5.1 Classical Vision Confirmed

The most striking feature of Jain and Krishna’s results is the inevitable evolution of the network toward greater complexity: starting from a sparse random graph, the production network evolves toward an ACS and then toward higher connectivity, specialisation, and interdependence. Technical change — the mutation of least-fit sectors — drives the production network toward the ever-evolving division of labour that Smith, Ricardo, and Marx identified as the defining dynamic of capitalist production. The ACS framework formalises this: the division of labour is not a contingent historical arrangement but the expected attractor of an autocatalytic system under competitive selection.

5.2 Relation to Thermodynamic Conservation

This paper isolates the *structural* level of the classical economic thermodynamics programme. The production network must be autocatalytic — must satisfy catalytic closure — as the precondition for any conservation law to hold. Without catalytic closure, the network decays toward entropy: sectors run down their capital stocks without replacement, the MELT conservation identity is violated, and value is not conserved in exchange. Conservation is an emergent property of ACS structure, not an assumption.

The maximum rate of profit R is the thermodynamic intensity of the ACS: how much catalytic surplus the self-reproducing network generates. When $R \rightarrow 0$ — as the dominant eigenvalue $\lambda_1 \rightarrow 1$ due to capital accumulation and rising organic composition — the network approaches the boundary of viability. This is the secular entropy maximum of the accumulation process, approached as the dominant ACS exhausts the productive surplus available for expansion.

5.3 Sraffian Prices and ACS Stability

The price system that supports the ACS in expanded reproduction is Sraffa’s prices of production p^* , which depend on both the rate of profit r and the production structure A :

$$p^* = \left(1 - \frac{r}{R}\right) v \left(I - \frac{r}{R} H\right)^{-1}$$

where v is the vector of labour values and $H = K(I - A)^{-1}$ is the capital matrix. As $r \rightarrow R$, p^* approaches the standard commodity proportions — the ACS attractor. The range of viable

prices collapses as $R \rightarrow 0$, which is the price-system expression of approaching the ACS viability boundary.

6 The Circuit Hamiltonian within the ACS Framework

The ACS structure of the Sraffian economy admits a natural thermodynamic operationalisation through the circuit Hamiltonian framework of Foley (1982, 1986), connecting catalytic productivity directly to labour-time accounting.

6.1 Capital in Labour-Time Units: $\mathcal{H} = K/\mu$

Define total capital $K = \sum_k K_k$ where $K_k = p_k A_k x_k$ is the monetary value of capital engaged in sector k . The **circuit Hamiltonian**

$$\mathcal{H} \equiv \frac{K}{\mu}$$

expresses total capital in abstract-labour-time units. $\mathcal{H}$ is the ACS's **total catalytic capacity measured in its thermodynamic substrate**: how many hours of living labour the production network represents at the current Monetary Equivalent of Labour Time $\mu = p^\top (I - A)x/\ell^\top x$.

In the ACS terms: $\mathcal{H}$ is proportional to the total “population” of the dominant ACS weighted by its labour-time cost of reproduction. When the ACS is fully intact (all basic commodities operating), $\mathcal{H}$ takes its maximum value for a given profit rate. When a keystone sector is disrupted, K falls discontinuously as capital in downstream sectors is devalued — and so does $\mathcal{H}$, by the same mechanism.

6.2 The Exploitation Rate ε as ACS Surplus Intensity

The **exploitation rate**

$$\varepsilon = \frac{\mu}{w} - 1 \quad \Longleftrightarrow \quad \mu = w(1 + \varepsilon)$$

where w is the money wage, measures how much the ACS produces above the reproduction requirements of labour-power. In ACS terms: $\varepsilon/(1 + \varepsilon)$ is the fraction of total living labour L that constitutes surplus labour S_ℓ — the labour-time the ACS generates above the threshold required to reproduce the workforce:

$$S_\ell = \frac{\varepsilon}{1 + \varepsilon} L, \quad V_\ell = \frac{1}{1 + \varepsilon} L, \quad L = V_\ell + S_\ell$$

The maximum rate of profit $R = 1/\lambda_1 - 1$ and the exploitation rate ε are related through the value composition of capital $\Omega_k = C_k/V_k$: at the sectoral profit rate $\rho_k = \varepsilon_k/(1 + \Omega_k)$, and at the uniform rate r for prices of production, $r \leq R$ with equality only at zero wages ($w = 0$). The dominant eigenvalue λ_1 is therefore a direct measure of the ACS's capacity to generate surplus above its own reproduction requirements — in labour-time units via ε , and in monetary units via R .

6.3 $\dot{\mathcal{H}}$ in ACS Terms

The time derivative of the circuit Hamiltonian, derived from the Foley stock equations:

$$\dot{\mathcal{H}} = p S_\ell = \frac{p \varepsilon}{1 + \varepsilon} L$$

where p is the capitalisation rate (fraction of surplus reinvested). In ACS terms this reads: **the ACS expands its total catalytic capacity at a rate equal to the fraction p of the surplus labour time S_ℓ that is recommitted to production** rather than consumed. This is the thermodynamic engine of accumulation: the second law (irreversible labour expenditure) generating surplus; the ACS structure determining how much of that surplus is reinvested in the network.

Simple reproduction ($p = 0$): $\dot{\mathcal{H}} = 0$ — the ACS reproduces at constant scale, its catalytic capacity unchanged. Expanded reproduction ($p > 0$): the ACS grows, adding catalytic links, deepening the division of labour, increasing connectivity — confirming the Jain-Krishna result that ACS networks evolve toward greater complexity.

6.4 Keystone Disruption, Phase Transition, and $\mathcal{H}$

When a keystone sector i is disrupted, the following chain of effects propagates through the Hamiltonian:

1. **ACS collapse:** the dominant ACS $\mathcal{B}$ loses nodes, reducing network connectivity
2. **Eigenvalue jump:** the effective dominant eigenvalue λ'_1 of the remaining network rises (fewer sectors reproduce as efficiently), so $R' = 1/\lambda'_1 - 1 < R$
3. **Profit rate compression:** with $R' < R$, the feasible range of r narrows; prices of production p^* must readjust
4. **Capital devaluation:** K falls as productive capital in downstream sectors is devalued — machines that cannot be operated without the keystone input lose value
5. **Hamiltonian collapse:** $\dot{\mathcal{H}} < 0$ — the circuit Hamiltonian falls discontinuously, not through the smooth non-conservation of expanded reproduction but through a structural phase transition in the ACS

This establishes the precise distinction between two modes of $\dot{\mathcal{H}} \neq 0$:

Mode	Mechanism	$\dot{\mathcal{H}}$	Character
Expanded reproduction	Surplus reinvestment ($p > 0$)	$+p\varepsilon L/(1 + \varepsilon)$	Smooth, continuous
Keystone crisis	ACS phase transition	Discontinuous drop	Sudden, structural

The extended correspondence table is now:

Sraffian / thermodynamic concept	ACS concept
$\mathcal{H} = K/\mu$	Total catalytic capacity in labour-time
$\varepsilon = \mu/w - 1$	ACS surplus intensity
$S_\ell = \varepsilon L/(1 + \varepsilon)$	Net catalytic surplus in labour-time
$\dot{\mathcal{H}} = pS_\ell$	Rate of ACS expansion (smooth)
$R = 1/\lambda_1 - 1$	Monetary expression of net catalytic surplus
Keystone disruption	Phase transition: $\dot{\mathcal{H}} \ll 0$ (discontinuous)
Long-wave: $R \rightarrow 0$	Secular entropy maximum: ACS approaches viability boundary

7 Conclusion

We have established the following correspondences between Sraffian production theory and auto-catalytic set theory:

Sraffian / thermodynamic concept	ACS concept
Basic commodities	Dominant ACS
Non-basic commodities	Peripheral (leaf) nodes
Standard commodity	Jain-Krishna attractor (leading eigenvector)
$R = 1/\lambda_1 - 1$	Net catalytic surplus (monetary)
$\varepsilon = \mu/w - 1$	ACS surplus intensity (thermodynamic)
$\mathcal{H} = K/\mu$	Total catalytic capacity in labour-time
$\dot{\mathcal{H}} = pS_\ell$	Rate of ACS expansion (smooth, expanded reproduction)
Keystone disruption	Phase transition: $\dot{\mathcal{H}}$ collapses discontinuously
$R \rightarrow 0$ (long wave)	Secular entropy maximum: ACS viability boundary
Long-wave recovery	ACS reconstitution around new attractor

The identification is exact, not analogical. The Sraffian economy *is* an ACS; the standard commodity *is* the Perron-Frobenius attractor; R *is* the net catalytic productivity; and the circuit Hamiltonian $\mathcal{H} = K/\mu$ is the ACS's total catalytic capacity in its thermodynamic substrate. Two modes of $\dot{\mathcal{H}} \neq 0$ are distinguished: the smooth continuous expansion of expanded reproduction ($\dot{\mathcal{H}} = pS_\ell > 0$) and the discontinuous collapse of keystone crisis (phase transition). Together they provide a complete thermodynamic account of capitalist accumulation and crisis grounded in the self-reproducing structure of the production network.

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